

# Smart Metro Ticket Booking System

## IDEATION PHASE

|               |                                   |
|---------------|-----------------------------------|
| Date          |                                   |
| Team ID       |                                   |
| Project Name  | Smart Metro Ticket Booking System |
| Maximum Marks | 4 Marks                           |

### 3. Brainstorming and Idea Prioritization

#### 3.1 Objective

A team brainstorming session was conducted to generate possible solution ideas for the problem of manual, queue-dependent metro ticket booking. Ideas were then grouped, evaluated, and prioritized based on feasibility on the ServiceNow platform, customer impact, and implementation effort.

#### 3.2 Raw Brainstormed Ideas

- Digital ticket booking portal built on ServiceNow Service Catalog with station and travel-date selection.
- Automated fare calculation engine based on source/destination distance and passenger count using Flow Designer.
- QR-code based digital ticket generation for entry/exit scanning at gates.
- SMS/Email/Push notifications for booking confirmation and ticket delivery.
- Wallet/UPI/card payment gateway integration for instant payment.
- Season pass / monthly pass booking and auto-renewal reminders.
- Live queue and crowd-status display integrated with the booking catalog item.
- Self-service ticket cancellation and refund workflow.
- Chatbot/Virtual Agent for booking assistance and FAQs.
- Admin dashboard for metro staff to monitor bookings, revenue, and station-wise traffic.
- Accessibility mode with voice assistance for elderly/differently-abled commuters.
- Group booking option for families or corporate travel.
- Integration with station gate hardware/IoT scanners for automatic QR validation.

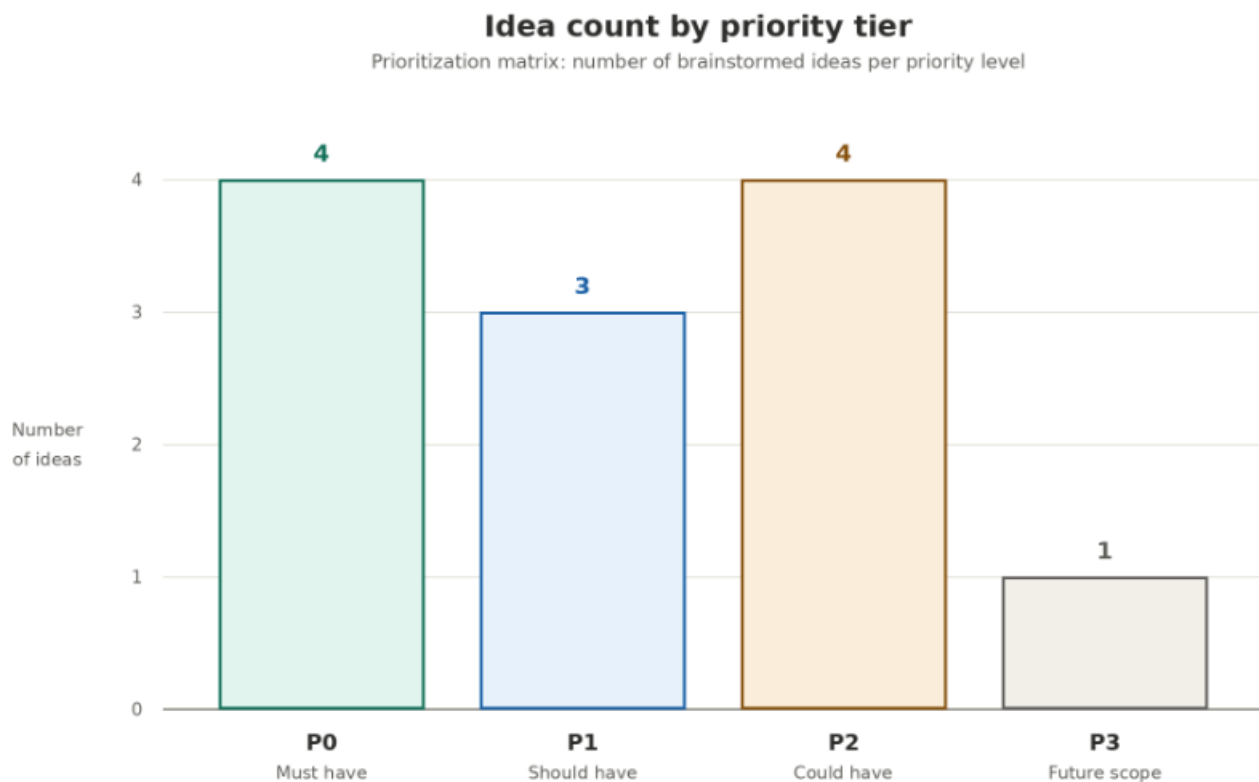
- Analytics and reporting module for peak-hour prediction and fare optimization.

### 3.3 Idea Grouping (Affinity Mapping)

| Theme                    | Grouped Ideas                                                          |
|--------------------------|------------------------------------------------------------------------|
| Core Booking Experience  | Service Catalog booking form, station/date/passenger selection         |
| Communication & Delivery | Notifications (SMS/Email/Push), booking confirmation                   |
| Payments                 | UPI/Card/Wallet integration, season pass renewal                       |
| Operations & Admin       | Admin dashboard, analytics/reporting, gate hardware integration        |
| Support & Accessibility  | Virtual Agent/chatbot, voice-assisted accessibility mode, cancellation |

### 3.4 Prioritization Matrix (Impact vs. Effort)

| Idea                                           | Customer Impact | Implementation Effort | Priority          |
|------------------------------------------------|-----------------|-----------------------|-------------------|
| Service Catalog based ticket booking form      | High            | Low                   | P0 - Must Have    |
| Automated fare calculation (Flow Designer)     | High            | Medium                | P0 - Must Have    |
| QR-code digital ticket generation              | High            | Low                   | P0 - Must Have    |
| Booking confirmation notifications (SMS/Email) | High            | Low                   | P0 - Must Have    |
| Payment gateway integration                    | High            | Medium                | P1 - Should Have  |
| Ticket cancellation/refund workflow            | Medium          | Medium                | P1 - Should Have  |
| Admin dashboard for staff/operations           | Medium          | Medium                | P1 - Should Have  |
| Season pass booking & renewal                  | Medium          | Medium                | P2 - Could Have   |
| Virtual Agent/chatbot support                  | Medium          | High                  | P2 - Could Have   |
| Gate hardware/IoT scanner integration          | Medium          | High                  | P2 - Could Have   |
| Accessibility voice-assistance mode            | Medium          | High                  | P2 - Could Have   |
| Analytics & peak-hour prediction               | Low             | High                  | P3 - Future Scope |



### 3.5 Finalized Ideas for Phase 1 (MVP)

Based on the prioritization matrix, the following ideas were selected as the Minimum Viable Product (MVP) scope for this project, all of which are achievable using core ServiceNow components:

- Service Catalog item for selecting source station, destination station, travel date, and number of passengers.
- Flow Designer workflow to automatically calculate the fare based on station distance/zone and passenger count.
- Automatic generation of a unique digital ticket with an embedded QR code upon successful booking.
- Notification module to instantly send the digital ticket and booking confirmation to the commuter.
- Basic booking history/status tracking for each user through the Service Catalog request record.